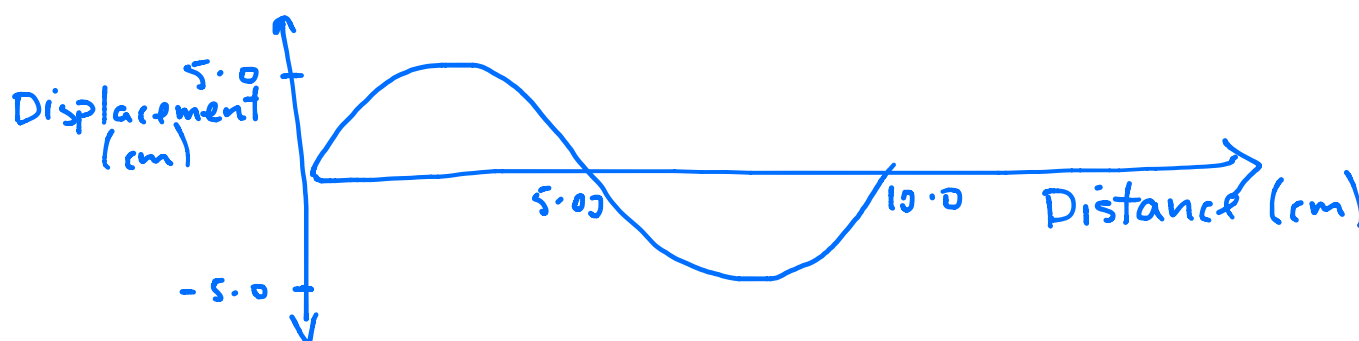


**YEAR 9 SCIENCE
PHYSICAL SCIENCES
REVISION - LIGHT**

1. (a) How do the particles move in a *transverse* wave? Give an example of this wave. *Up + down - perpendicular to direction of wave travel.*
- (b) How do the particles move in a *longitudinal* wave? Give an example of this wave. *Backwards + forwards - parallel to wave direction.*
- (c) Draw a displacement - distance graph to show the following. Make sure you give appropriate scales and labels.

amplitude = 5.00 cm
wavelength = 10.0 cm



2. (a) How would you describe **light** - what is it? *Pure energy*
- (b) Does light need a medium (substance) to travel in? How do you know? *No - light travels in space (a vacuum).*
3. (a) List the parts of the electromagnetic spectrum in order.
radio/TV wave, micro, infrared, visible, ultraviolet, X-ray, gamma ray
- (b) Give a use or characteristic for each part.
communication, cooking, heat, seeing, skin cancer, medicine, radiation
4. (a) What are the Laws of Reflection?
Angle of incidence = angle of reflection
- (b) Draw a simple diagram to show it.

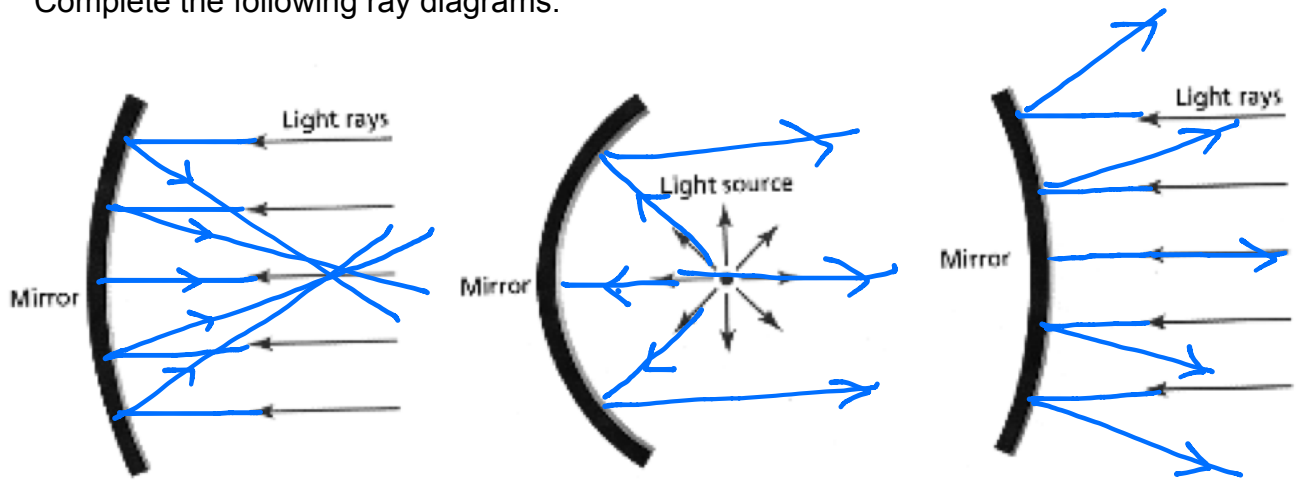


5. What is the difference between a **real image** and a **virtual image**?

Real - show on a screen.

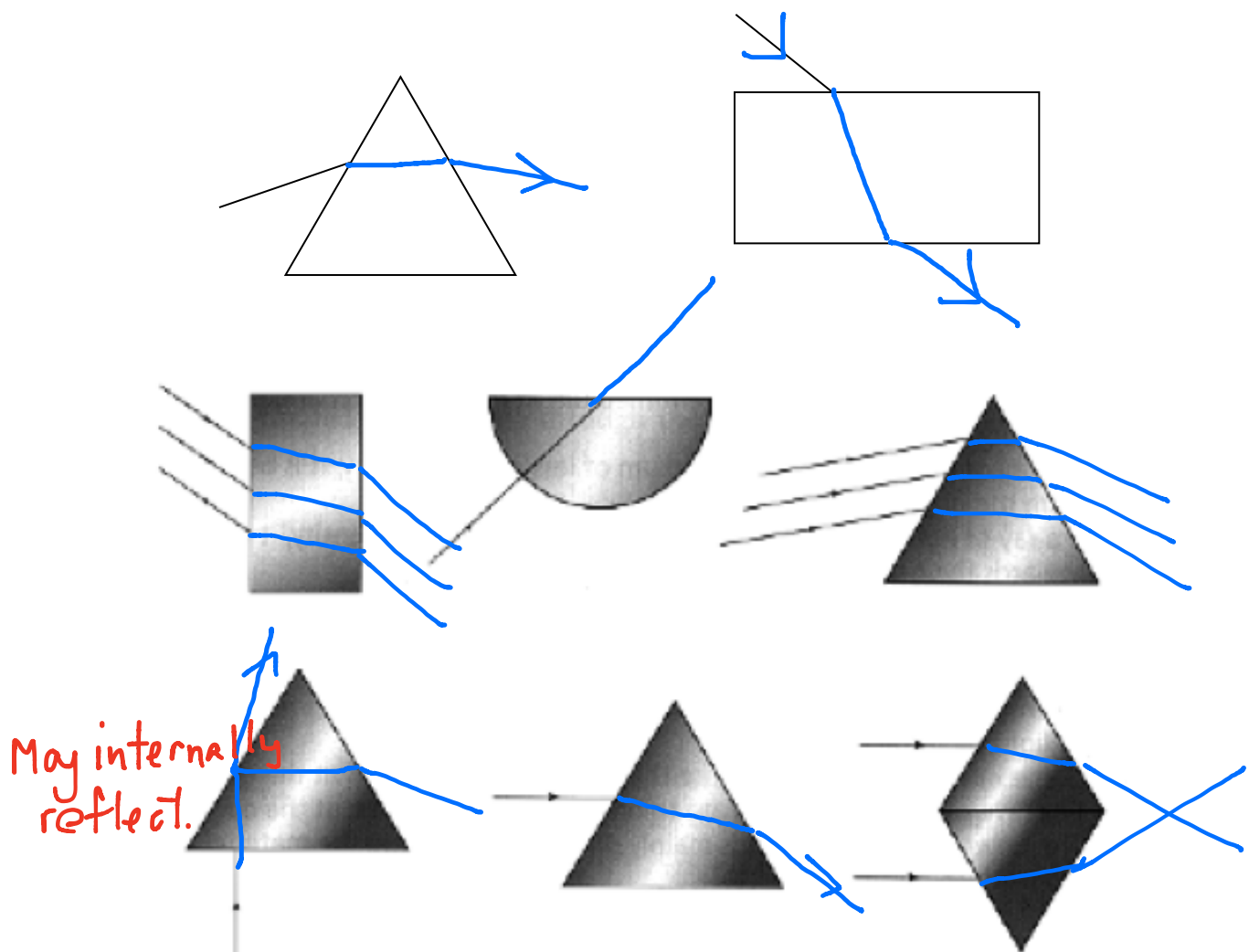
Virtual - can't show on a screen.

6. Complete the following ray diagrams.



7. (a) Define the term *refraction*.
Bending of light as it enters a new medium at an angle.
- (b) Will a light ray refract when it enters a new medium **at right angles** (along the normal)?
No - it goes in a straight line.
- (c) If a light ray enters a new medium where it *speeds up*, does it refract *towards* or *away from* the normal?
Away

8. Complete the following ray diagrams to show the path taken through the glass prisms.



9. (a) If a person on a jetty wishes to spear a fish swimming in the water (not directly below), where should they aim?

Below - the fish appears closer to the surface.

- (b) Draw a simple diagram to show this situation.



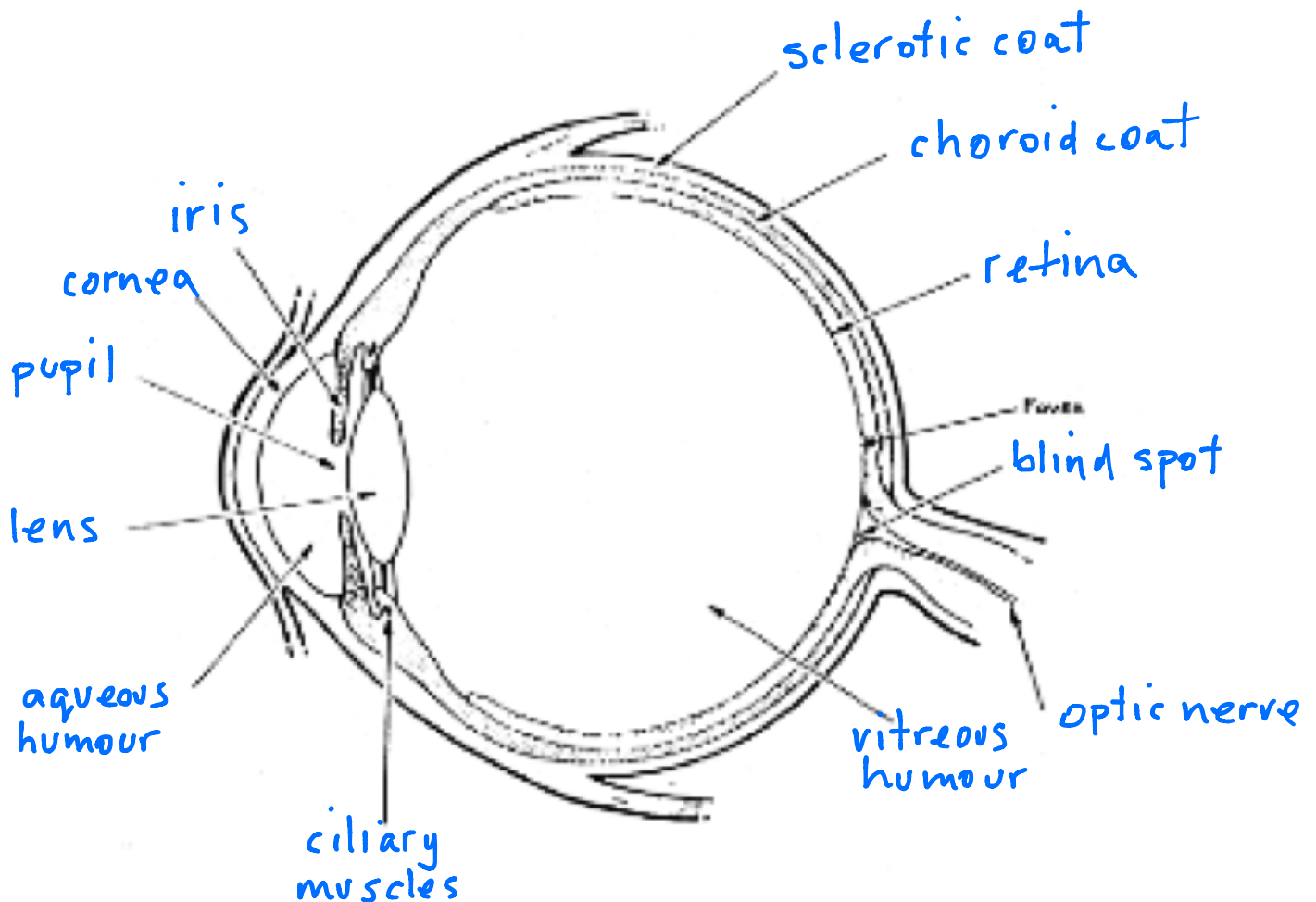
10. (a) If an object is placed a long way from a convex lens, what type of image is formed?

Upside down - real.

- (b) Where should an object be placed when a convex lens is used as a magnifying glass?

Close to the lens - inside focal point.

11. (a) Label the parts of the eye on the diagram below.



- (b) Give the major function for each part.

12. Describe how the lens of the eye changes to focus on a **close object** and a **distant object**. *Close- lens fat. Far away- lens thin.*

13. What is the **blind spot** in the eye?

Point where optic nerve meets the retina.